



COMPUTER ORGANISATION (TỔ CHỨC MÁY TÍNH)

WELCOME TO C.O.

Acknowledgement

- The contents of these slides have origin from School of Computing, National University of Singapore.
- We greatly appreciate support from Mr. Aaron Tan Tuck Choy for kindly sharing these materials.

Policies for students

- These contents are only used for students PERSONALLY.
- Students are NOT allowed to modify or deliver these contents to anywhere or anyone for any purpose.

Recording of modifications

- The list of the text books was changed (1-st book)
- Minor changes on online materials

COURSE DESCRIPTION

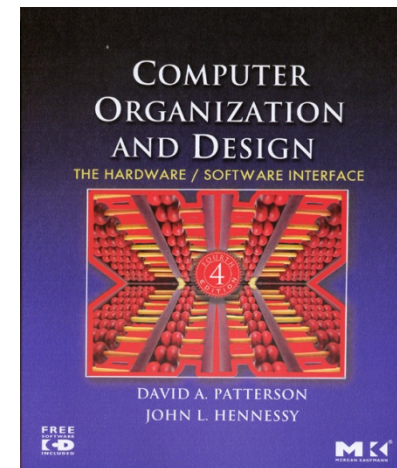
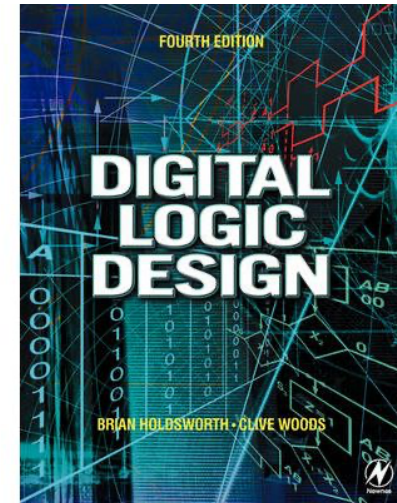
- The objective of this module is to familiarise students with the fundamentals of computing devices:
 - The basics of data representation
 - How the various parts of a computer work, separately and with each other
- **Topics:**
 - Data representation systems
 - Combinational and sequential circuit design
 - Assembly language
 - Processor execution cycles
 - Pipelining
 - Memory hierarchy
 - Input/output systems

SYLLABUS OUTLINE

- Number systems and codes
- Boolean algebra
- Logic gates and circuits
- Simplification
- Combinational circuits
- Sequential circuit
- Performance
- Assembly language
- The processor: Datapath and control
- Pipelining
- Memory hierarchy: Cache
- Input/output

TEXTBOOKS

- Brian Holdsworth, Clive Woods, [2002], **Digital Logic Design**, Elsevier Science & Technology Books (1-st part)
- David A. Patterson, John L. Hennessy, [2012], **Computer Organization and Design, Fourth Edition: The Hardware/Software Interface**, Morgan Kaufmann (2-nd part)



COURSE MATERIALS

You can find all lectures, tutorials, labs and solutions on Classroom or Library

ADMIN MATTERS

- Tutorials and labs will start in week 3

ASSESSMENT:

- 10% - Labs
- 20% - Mid-term exam
- 20% - Project
- 50% - Final exam

Q&A